

# The Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

*A more accurate surrogate can be a worse RL training environment — it exposes a rougher action→temperature surface that the policy exploits into failure on the real building. A role-separating hybrid keeps the surface smooth and the controller robust.*

## Surrogate training environment

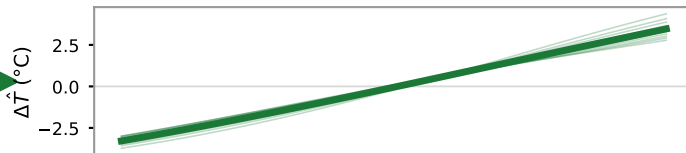
Coarse black-box v3  
(1 h step — less accurate)

Accurate twin  
(calibrated v3.5 / fine-res v3)

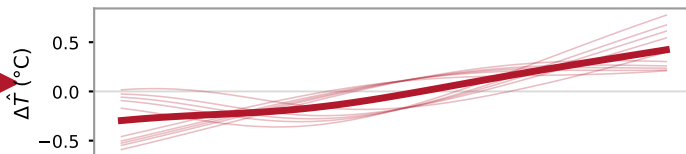
Hybrid  
(v3 rollout + frozen  
v3.5 censor)

## Action → next-temperature response surface (measured)

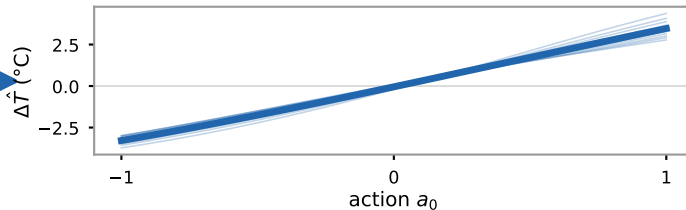
*smooth · monotone*



*rough · non-monotone*

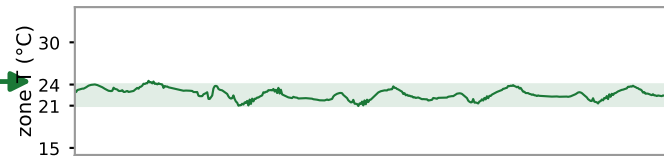


*smooth (v3 dynamics)  
+ plausibility censor*

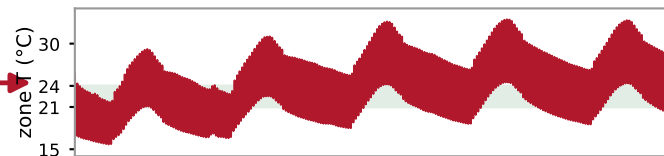


## Zone temperature on the live building (measured)

✓ **USABLE**



✗ **COLLAPSE**



✓ **ROBUST**

